

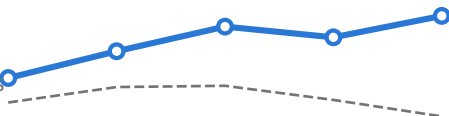
When should an iterative LLM loop stop?

Measure the utility available to signal-only stopping policies before building another gate.

1 Collect full-horizon loops

Code

repair | visible tests



Writing

revise | critic score



QA

retrieve | confidence



round 1

round T

hidden quality retained for offline replay

2 Decompose stopping regret

OOF
heuristic

hindsight
oracle



Task-clustered BCa endpoints | margins fixed by family

3 Test practical materiality

4 / 5 primary targets exclude a material effect

Code-8B total regret

remains inconclusive at the 0.02 margin

Predictive accuracy does not guarantee stopping utility.

Primary: $\lambda = 0.005$ | generation-only cost | stop-at-last